

A Packing of 9 Four-Spaces in $\mathbb{R}^{11}$

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Abstract

We give 9 4-dimensional subspaces of $\mathbb{R}^{11}$ whose smallest squared chordal distance is at least 2.863379. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.827986, obtained by deleting 1 of the 10 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 9 subspaces of dimension 4 in $\mathbb{R}^{11}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.863379 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 11$, $n = 4$. The cell $N = 9$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.827986, obtained by deleting 1 of the 10 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(11 - 4)/11 \cdot 9/8 = 2.863636$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m11n4N9.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
286337908879148573593801647282549216150040651888670034843491424390764199
240795645990853716947229688937555761186703525999891526563798644283600615
640110527094124302778408843447962219068805987209
```

denominator:

```
100000000000114459912529711451657858433120969172191910393517441298375991
216486251406783576139762327392440895923028159915064954072365765186465409
555016175666362261387078360815208322972358191205
```

B Coordinates

9 blocks of 11 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m11n4N9.txt`.

```
494297559289 -47470331358 -374896648458 354019670463758177939 229937276057 -107077506477 399588588223 159148478016 -265317103458 -365367517493 149464278417
373648564337 -168190246358 526187270249 -13747142246374575030448 412195316517 4381701825 173907247109 24818899897 -585027393042 -336521437383 12622428308
-144778679248 43315502168 277041293420 455567091610-18608333847 85684223299 -415524690738 -43269585711284975877573 -351380588492 -284172049698 260695634660
61018154086 -148252057146 -319187120464 -552324695927077382618 162306498707 49148067188 486014104225 -117557906116 213189103983 -247950235599 290884885009
220112709133 -489665152395 22834682130 -16336417137304172079057 -378576287810 327738465827 188907133632277391031 148365085367 221716742677 585337160023
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-217715553175 585912332401 -226813328148 -944841830467102227597 -48337733208 -713961366579 251516795280204034110759 -160792693475 309436817966 455142152306
-127647846248 -264985106424 -221432888511 13961437932319792163 24117045161 317310863358 -334763032660148336837215 -468696310501 277275717879 -74217537537
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```